How-To Library

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Configuring Azure Active Directory Authentication for Microsoft Azure SQL Data Warehouse Connector

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Abstract

This article explains how to configure Azure Active Directory (AAD) authentication when you use Microsoft Azure SQL Data Warehouse Connector to connect to Microsoft Azure SQL Data Warehouse. By default, the Microsoft Azure SQL Data Warehouse connection uses Microsoft SQL Server authentication.

Supported Versions

- Microsoft Azure SQL Data Warehouse Connector

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Overview

Sales department of your organization has a large amount of customer details for all regions in flat files. The sales analyst needs to analyze the customer data in a given region in a short span of time. You need to write flat files data to Microsoft Azure SQL Data Warehouse tables. Microsoft Azure SQL Data Warehouse has been configured to use Azure Active Directory (AAD) user accounts for secured access.

Microsoft Azure SQL Data Warehouse Connector enables you to configure AAD authentication to connect to Microsoft Azure SQL Data Warehouse. Default is Microsoft SQL Server authentication.

To configure AAD authentication, perform the following tasks:

1. Import server certificate.
2. Add an active directory administrator account to manage the Microsoft Azure SQL Data Warehouse.
3. Create a user.
4. Configure connection properties.

Import Server Certificate

If a trust store file is not configured for your organization and you want to use AAD authentication with Active Directory Federation Services in Azure, import the server certificate to the `cacerts` file at the following location:

`<Secure Agent installation directory>\jre\lib\security` file.

Use the following command to import the server certificate:

```
keytool -import -trustcacerts -alias <alias name of the certificate> -file <certificate file path> -keystore <Secure Agent installation directory>\jre\lib\security\cacerts -storepass <password for the truststore>
```

Note: If Active Directory Federation Services is used for AAD authentication, you cannot run a task on a Hosted Agent,

Add an Active Directory Admin Account

Perform the following steps to add an administrator to manage Microsoft SQL Server system that hosts the Microsoft Azure SQL Data Warehouse:

1. Log in to the Microsoft Azure portal using your credentials. The Dashboard page appears.
2. From the All Resources page, select the Microsoft SQL Server system that hosts the Microsoft Azure SQL Data Warehouse.
3. Select the **Active Directory admin** option under the settings displayed for the selected Microsoft SQL Server system.
4. Click **Set admin**. The **Add admin** page appears.
5. Enter the email ID that you want to use for the administrator account and click **Select**.
6. Click **Save**.

Create a User

Perform the following steps to create a user:

1. Use the active directory administrator account to connect to the Microsoft Azure SQL Data Warehouse. You can use Microsoft SQL Server Management Studio to connect to the Microsoft Azure SQL Data Warehouse.
2. Type and run the following command to create a user:
   ```
   create user [user@foobar.com] from external provider;
   ```
   For more information, see Microsoft Azure documentation.
3. Assign the required privileges to the user.

Configure Microsoft Azure SQL Data Warehouse Connection Properties

When you create a Microsoft Azure SQL Data Warehouse connection, configure AAD user name and JDBC connection string with AAD authentication in connection properties.

The following table lists the connection properties to enable AAD authentication:

Connection property	Description
Azure DW JDBC URL	Microsoft Azure Data Warehouse JDBC connection string. Example for Azure Active Directory (AAD) authentication: `jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30;Authentication=ActiveDirectoryPassword;` The default authentication is Microsoft SQL Server authentication.
Azure DW JDBC Username	User name to connect to the Microsoft Azure SQL Data Warehouse account. For AAD authentication, provide the AAD user name that you created in the "Create a User" on page 3 section.
Azure DW JDBC Password	Password to connect to the Microsoft Azure SQL Data Warehouse account.

Authors

Madhuri Velluri

Chanchal Das